In 1960, Leonard Schiff of Stanford University and George Pugh of the U.S. Department of Defense independently suggested that general relativity's prediction of frame dragging might be experimentally tested using the rotational motion of the earth. Schiff and Pugh realized that according to Newtonian physics, a spinning gyroscope—a spinning wheel that's attached to an axis—floating in orbit high above the earth's surface would point in a fixed and unchanging direction. But, according to general relativity, its axis would rotate ever so slightly because of the earth's dragging of space. Since the earth's mass is puny in comparison with the hypothetical hollow sphere used in the Pfister and Braun calculation above, the degree of frame dragging caused by the earth's rotation is tiny. The detailed calculations showed that if the gyroscope's spin axis were initially directed toward a chosen reference star, a year later, slowly swirling space would shift the direction of its axis by about a hundred-thousandth of a degree. That's the angle the second hand on a clock sweeps through in roughly two millionths of a second, so its detection presents a major scientific, technological, and engineering challenge.

Four decades of development and nearly a hundred doctoral dissertations later, a Stanford team led by Francis Everitt and funded by NASA is ready to give the experiment a go. During the next few years, their *Gravity Probe B* satellite, floating 400 miles out in space and outfitted with four of the most stable gyroscopes ever built, will attempt to measure frame dragging caused by the earth's rotation. If the experiment is successful, it will be one of the most precise confirmations of general relativity ever achieved, and will provide the first direct evidence of a Machian effect.[2] Equally exciting is the possibility that the experiments will detect a deviation from what general relativity predicts. Such a tiny crack in general relativity's foundation might be just what we need to gain an experimental glimpse into hitherto hidden features of spacetime.

Catching the Wave

An essential lesson of general relativity is that mass and energy cause the fabric of spacetime to warp; we illustrated this in Figure 3.10 by showing the curved environment surrounding the sun. One limitation of a still